

README — dsmz_tcga_correlation.R

Purpose

This script computes *Spearman* gene-expression correlations between DSMZ cancer cell lines and TCGA patient tumors. It harmonizes gene IDs, optionally adjusts TCGA expression for tumor purity, summarizes best TCGA cohort matches per cell line (overall and organ-constrained), and produces diagnostic plots and joint UMAP embeddings (TCGA cohort means + DSMZ samples).

Inputs (edit at top of the R script)

TCGA_SE_RDS: SummarizedExperiment RDS with raw/counts (genes × samples).

DSMZ_RDS: Data frame with columns `Ensembl_ID`, `gene_name`, optional `Ensembl_ID_with_version`, then sample columns.

DSMZ_META_CSV: Metadata with a `sample_name` column (exactly matching DSMZ column names).

PURITY_TSV (optional): Two-column table `sample`, `purity` for TCGA samples.

Outputs (written to OUTDIR)

Tables:

- `spearman_all_wide.csv` — DSMZ × TCGA-cohort matrix of Spearman ρ .
- `spearman_all_long.csv` — long-form correlations (`sample_id`, `tcga_cohort`, ρ).
- `spearman_all.rds` — full correlation matrix (RDS).
- `spearman_best_by_sample.csv` — best overall match and organ-constrained match per DSMZ sample.
- `dsmz_tcga_mapping.csv` — DSMZ sample → assigned TCGA cohort (organ-constrained).
- `umap_tcga_means_plus_dsmz.csv` — 2D embedding coordinates and labels.

Plots (PDF):

- `violin_rho_by_organ.pdf` — best overall ρ grouped by DSMZ organ.
- `heatmap_top25_dsmz_vs_tcga_cohorts.pdf` — top 25 DSMZ samples vs. TCGA cohorts.
- `organ_tcga_assignment_counts.pdf` — counts heatmap of DSMZ organ × assigned TCGA cohort.
- `umap_tcga_means_plus_dsmz.pdf` — joint UMAP colored by source (TCGA/DSMZ).
- `umap_dsmz_by_organ.pdf` — DSMZ colored by organ; TCGA means in gray.

Debug/Provenance:

- `head_tcga_genes.txt`, `head_dsmz_genes.txt` — quick gene ID snapshots.
- `common_genes_head.txt` — first shared genes after harmonization.
- `sessionInfo.txt` — R session details.

Pipeline Overview

1) **Load & sanity checks** — Confirms required columns and prints dimensions.

- 2) Harmonize gene IDs** — Removes Ensembl version suffix from TCGA (e.g., .15) and collapses duplicates; DSMZ is keyed on version-less Ensembl_ID.
- 3) Build DSMZ matrix** — Coerces sample columns to numeric, drops all-NA sample columns, aligns to metadata.
- 4) Organ labels** — Derives broad organ from Origin via regex rules.
- 5) Intersect genes** — Uses *all* valid shared genes (no reduction by default).
- 6) Tumor purity (optional)** — Removes purity-anti-correlated genes ($FDR < 0.01$ & $\rho < -0.4$) and regresses out infiltration ($1 - \text{purity}$) with limma.
- 7) TCGA cohort means** — Selects label from project_id/study/disease_type, computes per-cohort mean expression.
- 8) Variable gene set (optional)** — tcga_5k, tcga_10k, or all (IQR ranking).
- 9) Spearman correlations** — DSMZ samples $\times$ TCGA cohorts on selected genes.
- 10) Organ-constrained mapping** — Restricts candidate cohorts by DSMZ organ and picks max ρ .
- 11) Plots** — Violin, heatmaps, and joint UMAP visualizations.
- 12) UMAP** — Per-gene z-scored matrix of TCGA cohort means + DSMZ samples (cosine metric).

Key Configuration Flags

VAR_GENE_SET: "tcga_5k" | "tcga_10k" | "all" (controls optional variable-gene selection).

MIN_SHARED: Warns if shared genes below this threshold (default 1000).

PURITY_TSV: Provide external purity estimates if not embedded in colData(tcga_se).

OUTDIR: All results are written here; created if missing.

Usage

Run the script non-interactively:

```
Rscript dsmz_tcga_correlation.R
```

Or submit via SLURM (example):

```
#!/bin/bash #SBATCH --job-name=dsmz_tcga_corr #SBATCH
--output=/path/to/logs/dsmz_tcga_corr_%j.out #SBATCH
--error=/path/to/logs/dsmz_tcga_corr_%j.err #SBATCH --cpus-per-task=8 #SBATCH
--mem=48G #SBATCH --time=24:00:00 module load R/4.3 # or activate your conda
env Rscript /path/to/dsmz_tcga_correlation.R
```

Troubleshooting

Few shared genes — Your TCGA rows may still contain versioned Ensembl IDs (e.g., ENSG... .15). The script strips versions and collapses duplicates; verify head_tcga_genes.txt vs. head_dsmz_genes.txt.

Non-numeric columns — DSMZ sample columns are coerced with as.numeric; all-NA columns get dropped and reported.

Purity adjustment skipped — Provide a purity column in colData or a PURITY_TSV file.

UMAP errors — Ensure uwot is installed and there are enough common genes after filtering.

Notes & Assumptions

- Correlations use *Spearman* with use = 'pairwise.complete.obs'.
- Cohort labels prefer project_id then study then disease_type.
- Organ-to-TCGA mapping is heuristic and can be edited in the script (organ2tcga).
- All numeric operations run on double-precision matrices (storage mode enforced).

References

- TCGA Pan-Cancer Atlas resources; Broad GDAC Firehose / GDC data portal.
- DSMZ Cell Line Services documentation for lineage annotations.
- Law et al., *limma* for differential expression; utilities used for regression.
- McInnes et al., UMAP: Uniform Manifold Approximation and Projection.